

Shahista Tamkeen

Machine Learning Engineer

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Professional Summary

Machine Learning Engineer with 3+ years of experience designing, building, and deploying production AI applications across fraud detection, predictive analytics, Generative AI, and intelligent automation. Experienced in developing scalable machine learning models, Retrieval-Augmented Generation (RAG) applications, Graph Neural Networks (GNNs), and cloud-native MLOps solutions using Python, PyTorch, Apache Spark, LangChain, LangGraph, and Microsoft Azure. Strong software engineering background with expertise in backend development, REST APIs, distributed data processing, and containerized deployments, delivering reliable, explainable, and production-ready AI solutions for enterprise applications.

Technical Skills

Programming Languages: Python, Java, SQL, Bash

AI, Machine Learning & Deep Learning: Scikit-learn, XGBoost, LightGBM, CatBoost, TensorFlow, PyTorch, PyTorch Geometric (GNNs), CNNs, RNNs, LSTMs, Transformers, Supervised Learning, Unsupervised Learning, Feature Engineering, Hyperparameter Tuning, Model Evaluation, Explainable AI (SHAP, LIME)

Generative AI & LLMs: Retrieval-Augmented Generation (RAG), LangChain, LangGraph, Azure OpenAI, OpenAI API, AI Agents, Prompt Engineering, Function Calling, Semantic Search, Vector Databases, Pinecone, Llama, Mistral

Software Engineering & Backend Development: Object-Oriented Programming (OOP), Data Structures & Algorithms, FastAPI, Flask, REST APIs, Microservices, Apache Kafka, Design Patterns, Git, GitHub, CI/CD, Unit Testing, Integration Testing

Data Engineering: Apache Spark, Spark Structured Streaming, ETL Pipelines, Snowflake, SQL Server, Pandas, NumPy

MLOps & Deployment: MLflow, Docker, Kubernetes, FastAPI Model Serving, Model Monitoring, GitHub Actions, CI/CD Pipelines

Cloud & Databases: Microsoft Azure, AWS, PostgreSQL, SQL Server, Redis, Pinecone

Data Visualization: Power BI, Tableau, Streamlit, Matplotlib

Professional Experience

AI / Machine Learning Engineer | AIG | USA

Jan 2026 - Present

- Developed an XGBoost-based fraud detection solution using Python and Apache Spark to identify high-risk payment transactions, improving fraud detection accuracy by 22% across enterprise financial datasets.
- Engineered Graph Neural Network (GNN) models with PyTorch Geometric to analyze relationships between customers, merchants, and devices, uncovering complex fraud patterns that improved detection accuracy by 18%.
- Built a Retrieval-Augmented Generation (RAG) platform using LangChain, LangGraph, Azure OpenAI, and Pinecone, enabling investigators to retrieve policy and fraud intelligence faster and reduce investigation time by 30%.
- Designed FastAPI-based inference services that exposed machine learning models through scalable REST APIs, simplifying integration with enterprise fraud detection applications.
- Standardized feature engineering, experiment tracking, and model validation workflows with MLflow, improving reproducibility and accelerating model deployment across multiple projects.
- Containerized machine learning services with Docker and Kubernetes to support reliable deployments, scalable inference, and consistent application releases across environments.
- Partnered with software engineers, data engineers, and fraud analysts throughout the development lifecycle to deliver explainable AI solutions while following Git workflows, code reviews, and CI/CD best practices.

Machine Learning Engineer | Persistent Systems | India

Nov 2021 - Jul 2024

- Developed machine learning models for credit risk assessment, fraud detection, and customer analytics using Python, Scikit-learn, XGBoost, and LightGBM, improving lending risk prediction accuracy by 24%.
- Built scalable Spark and SQL data pipelines that processed over 5 million financial records, reducing preprocessing time by 30% and improving data readiness for model training.
- Designed and deployed FastAPI-based REST services to integrate machine learning models into enterprise banking applications, enabling real-time risk scoring for business users.
- Improved model performance through feature engineering, hyperparameter optimization, cross-validation, and SHAP/LIME explainability, increasing prediction precision by 21% while improving model transparency.
- Automated model training and deployment workflows using Mlflow, Docker, and Apache Airflow, reducing manual operational effort and improving release consistency.
- Worked closely with product managers, business analysts, and engineering teams to translate business requirements into production-ready machine learning solutions that supported lending and fraud prevention initiatives.
- Created Power BI and Tableau dashboards to visualize fraud trends, customer segmentation, and model performance, helping business stakeholders monitor KPIs and make data-driven decisions.

Education

Master of Science in Computer Science

Elmhurst University, Elmhurst, IL | Aug 2024 – May 2026

Projects

Digital Twin Multi-Agent AI Platform

- Designed and developed a production-ready multi-agent AI platform using LangGraph, LangChain, and Azure OpenAI to automate career guidance, financial planning, and personalized learning workflows.
- Implemented Retrieval-Augmented Generation (RAG), semantic search, vector databases, and agent orchestration to deliver contextual, multi-step AI responses with improved accuracy and reliability.
- Built scalable RESTful APIs using FastAPI and integrated PostgreSQL for persistent memory, user management, and secure backend services supporting concurrent AI workflows.
- Technologies: Python, LangGraph, LangChain, Azure OpenAI, FastAPI, PostgreSQL, React, Docker

AI-Powered 3D Point Cloud Quality Assessment System

- Developed an AI-powered computer vision application for automated quality assessment of construction point cloud datasets using Open3D and geometric analysis techniques.
- Implemented point cloud registration, deviation analysis, and defect detection algorithms to improve inspection accuracy and reduce manual quality verification efforts.
- Built an interactive visualization platform using React and Three.js for real-time 3D rendering, engineering model comparison, and defect analysis.
- Technologies: Python, Open3D, Computer Vision, FastAPI, React, Three.js

Brain Age Prediction Using Deep Learning

- Developed a deep learning framework using Convolutional Neural Networks (CNNs) to predict biological brain age from MRI-based neuroimaging datasets.
- Built an end-to-end pipeline including image preprocessing, augmentation, feature extraction, model training, and evaluation to support medical imaging research.
- Published the research in the International Journal of Engineering Research (IJER), demonstrating the application of deep learning techniques in healthcare.
- Technologies: Python, TensorFlow, CNN, Deep Learning, Medical Imaging

Customer Risk & Loan Prediction

- Developed supervised machine learning models to predict customer credit risk and loan approval outcomes using structured financial datasets.
- Performed feature engineering, exploratory data analysis, and model optimization to improve prediction performance and support lending decisions.
- Built interactive dashboards using Streamlit and Power BI to visualize customer segmentation, credit risk, and key business insights for decision-makers.
- Technologies: Python, Scikit-learn, SQL, Streamlit, Power BI

Certifications

- Google Cloud Certification – Google Cloud
- Microsoft Certified: Azure Fundamentals (AZ-900) – Microsoft
- AI & Data Science Certification – IBM
- Java Programming Certification – Oracle
- C Programming Certification (CLA)

Publications

AI for Brain Age Prediction Using Deep Learning
International Journal of Engineering Research (IJER)

- Published research on deep learning-based brain age prediction using MRI neuroimaging datasets.
- Developed a Convolutional Neural Network (CNN) framework for image preprocessing, feature extraction, model training, and biological brain age estimation.